Name _____ Period _____ Date _____

7-2 Lesson Quiz: Evaluating Trigonometric Functions

short form questions keep their Savvas numbers (1, 2, 4); show your work

1. What are the sine and cosine of $\frac{2\pi}{3}$? Write one tile in each box.

$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\sqrt{3}$	4

$\sin\frac{2\pi}{3} =$ ☐ $\cos\frac{2\pi}{3} =$ ☐

Work:

2. What is $\sin\theta$ if $\cos\theta = \frac{12}{13}$ and θ is in Quadrant IV?

☐ **A.** $\sin\theta = \frac{12}{13}$

☐ **B.** $\sin\theta = -\frac{5}{13}$

☐ **C.** $\sin\theta = \frac{5}{13}$

☐ **D.** $\sin\theta = -\frac{12}{13}$

Work:

4. Which of the following are true? Select all that apply.

☐ **A.** $\csc 225° = -\sqrt{2}$

☐ **B.** $\cot 225° = \sqrt{2}$

☐ **C.** $\cot 225° = 1$

☐ **D.** $\sec 225° = 1$

☐ **E.** $\sec 225° = -\sqrt{2}$

☐ **F.** $\csc 225° = \sqrt{2}$

Work:

Scoring: Question 1 and Question 2 are 10 points each. Question 4 is a bonus of up to 5 points. Total out of 20.

7-2 Lesson Quiz, short form: answer key

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1. $\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$**,** $\cos\frac{2\pi}{3} = -\frac{1}{2}$**.** Quadrant II, reference angle $\pi - \frac{2\pi}{3} = \frac{\pi}{3}$; terminal point $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. Distractors: $\frac{1}{2}$ and $-\frac{\sqrt{3}}{2}$ come from using the wrong leg (reference angle $\frac{\pi}{6}$ instead of $\frac{\pi}{3}$, the $\frac{5\pi}{6}$ point); $\sqrt{3}$ is $|\tan\frac{2\pi}{3}|$. Savvas quiz item 1, DOK 1, Skills Review and Practice F84. Taught by Example 4C (the same angle), so this item is a recall check of the packet. **10 points:** 5 per correct box.

2. **B,** $\sin\theta = -\frac{5}{13}$**.** $\sin^2\theta + \left(\frac{12}{13}\right)^2 = 1$, so $\sin^2\theta = 1 - \frac{144}{169} = \frac{25}{169}$ and $\sin\theta = \pm\frac{5}{13}$; in Quadrant IV the y-coordinate is negative, so $\sin\theta = -\frac{5}{13}$. Distractors: C has the wrong sign; A and D reuse 12 (the cosine's numerator) instead of solving the identity. Savvas quiz item 2, DOK 1, F84 and F92. Taught by Example 6B (the Pythagorean Identity, then the quadrant picks the sign); Try It 6a is the Quadrant IV parallel. **10 points:** B 10; C (right size, wrong sign) 5 when the work shows $\sin^2\theta = \frac{25}{169}$; 0 otherwise.

4. **A, C, E.** 225° is in Quadrant III with reference angle 45°; terminal point $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$, the same point as Example 4A ($\frac{5\pi}{4} = 225°$). $\csc 225° = \frac{1}{-\sqrt{2}/2} = -\frac{2}{\sqrt{2}} = -\sqrt{2}$; $\sec 225° = -\sqrt{2}$; $\cot 225° = \frac{x}{y} = 1$. B, D and F are false (F has the wrong sign). Savvas quiz item 4, DOK 1, F84. Taught by Example 4A's terminal point and the packet's reciprocal box (Example 5, cut, holds the reciprocal definitions); the packet Practice (135°, TE Practice 14 and Do You Know How 11) is the parallel. **Bonus 5:** A, C and E and no other box; 3 for two of the three and no wrong box; 0 otherwise.

Items left off this form (available for a retake or Form B): Savvas item 3, "Select all the correct statements" about $\tan\left(-\frac{4\pi}{3}\right)$, $\cos\left(-\frac{7\pi}{4}\right)$, $\cot\left(-\frac{11\pi}{3}\right)$, $\sec\frac{7\pi}{6}$, $\sin\frac{13\pi}{6}$ (answer A, D, E; DOK 1, F84; Example 4 with the coterminal step from Lesson 7-1 and the reciprocals of cut Example 5: A, $-\frac{4\pi}{3} + 2\pi = \frac{2\pi}{3}$, tan = $-\sqrt{3}$; D, $\cos\frac{7\pi}{6} = -\frac{\sqrt{3}}{2}$, sec = $-\frac{2\sqrt{3}}{3}$; E, $\frac{13\pi}{6} - 2\pi = \frac{\pi}{6}$, sin = $\frac{1}{2}$; B and C have the wrong sign); Savvas item 5, Melissa's two-way radio, 8-mile radius, Oscar 30° west of south (answer $\left(-4, -4\sqrt{3}\right)$: standard-position angle 240°, $8\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$; DOK 2, F84; any circle centered at the origin, cut Example 7). Item 5's figure (a radius-8 circle on compass axes, grid every 2, a radius 30° from the south axis) could be redrawn; it is left off because no kept Example teaches it, not because of the figure.

IXL replacement: SmartScore 70 or above on 6H8, XJJ and PAE replaces this quiz with 20/20.

Teacher notes. Every item on the Savvas quiz is DOK 1 except item 5 (DOK 2); the 7-2 TE prints no DOK 3 item anywhere. Item 2 is the Pythagorean Identity of Example 6, the same skill as Topic 7 test items 1, 3 and 16. The TE key for Try It 4 prints $\tan\frac{4\pi}{3} = \frac{\sqrt{3}}{3}$; the correct value is $\sqrt{3}$. The TE key for Practice 14 prints $\sec 135° = \sqrt{2}$; the correct value is $-\sqrt{2}$.